Configuring a Single-Area OSPFv3-based IPv6 Network

Difficulty (out of 4 stars): ★★★★ Recommended study time: 3 hours

Contents

1 About This Lab

1.1 Task Description

Technological changes require migrating an IPv4 enterprise network to IPv6. Your first step as an administrator is to design and reconstruct the current network to support IPv6. In this lab, configure an OSPFv3-based IPv6 network with a single area.

After completing this task, you will be able to independently set up and configure a small IPv6 network using OSPFv3.

1.2 Objectives

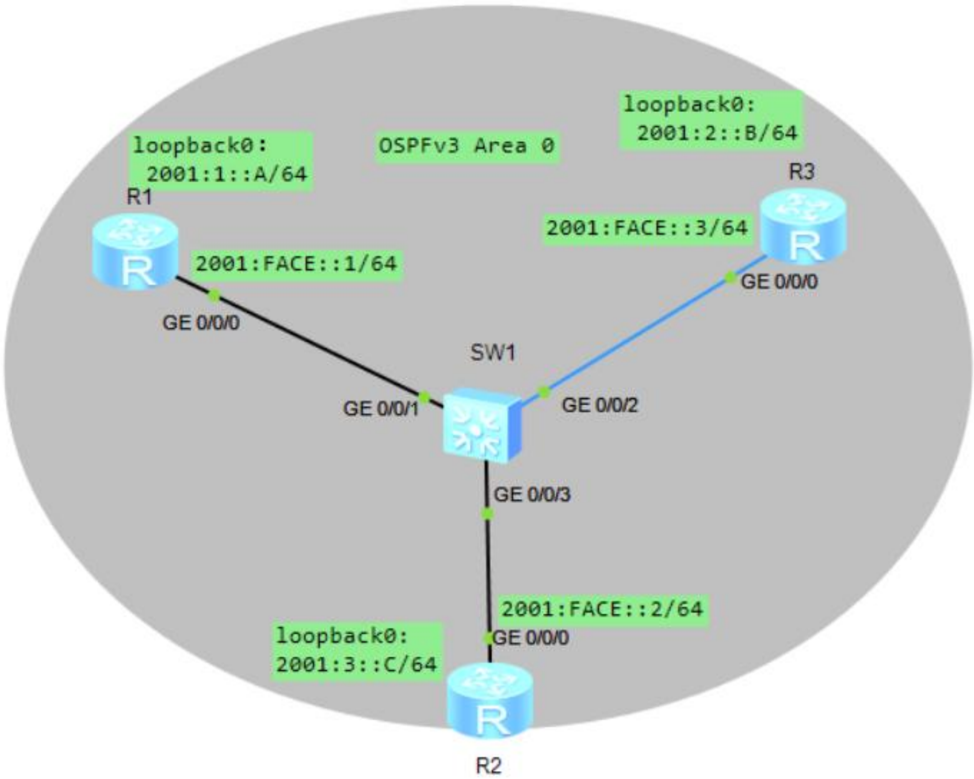
On completing this task, you will be able to:

1. Configure IPv6 addresses.
2. Configure OSPFv3.
3. Be familiar with OSPFv3 display commands.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology



2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]ipv6
[R1]interface loopback 0
[R1-LoopBack0]ipv6 enable
[R1-LoopBack0]ipv6 address 2001:1::A 64
[R1-LoopBack0]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ipv6 enable
[R1-GigabitEthernet0/0/0]ipv6 address fe80::1 link-local
[R1-GigabitEthernet0/0/0]ipv6 address 2001:FACE::1 64
[R1-GigabitEthernet0/0/0]quit
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]ipv6
[R2]interface loopback 0
[R2-LoopBack0]ipv6 enable
[R2-LoopBack0]ipv6 address 2001:2::B 64
[R2-LoopBack0]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]ipv6 enable
[R2-GigabitEthernet0/0/0]ipv6 address fe80::2 link-local
[R2-GigabitEthernet0/0/0]ipv6 address 2001:FACE::2 64
[R2-GigabitEthernet0/0/0]quit
[R2]
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]ipv6
[R3]interface loopback 0
[R3-LoopBack0]ipv6 enable
[R3-LoopBack0]ipv6 address 2001:3::C 64
[R3-LoopBack0]
[R3-LoopBack0]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]ipv6 enable
[R3-GigabitEthernet0/0/0]ipv6 address fe80::3 link-local
[R3-GigabitEthernet0/0/0]ipv6 address 2001:FACE::3 64
[R3-GigabitEthernet0/0/0]quit
[R3]
```

3 Task Implementation

3.1 Checking Basic Configurations

Run the **display ipv6 interface brief** command on R1, R2, and R3 to check interface configurations.

```
[R1]display ipv6 interface brief
*down: administratively down
(l): loopback
(s): spoofing
Interface                    Physical              Protocol
GigabitEthernet0/0/0         up                    up
[IPv6 Address] 2001:FACE::1
LoopBack0                    up                    up(s)
[IPv6 Address] 2001:1::A
[R1]

[R2]display ipv6 interface brief
*down: administratively down
(l): loopback
(s): spoofing
Interface                    Physical              Protocol
GigabitEthernet0/0/0         up                    up
[IPv6 Address] 2001:FACE::2
LoopBack0                    up                    up(s)
[IPv6 Address] 2001:2::B
[R2]

[R3]display ipv6 interface brief
*down: administratively down
(l): loopback
(s): spoofing
Interface                    Physical              Protocol
GigabitEthernet0/0/0         up                    up
[IPv6 Address] 2001:FACE::3
LoopBack0                    up                    up(s)
[IPv6 Address] 2001:3::C
[R3]
```

Run the **ping** command on R1 to check the connectivity between devices.

```
[R1]ping ipv6 -c 2 2001:FACE::2
  PING 2001:FACE::2 : 56   data bytes, press CTRL_C to break
    Reply from 2001:FACE::2
    bytes=56 Sequence=1 hop limit=64   time = 110 ms
    Reply from 2001:FACE::2
    bytes=56 Sequence=2 hop limit=64   time = 50 ms

  --- 2001:FACE::2 ping statistics ---
    2 packet(s) transmitted
```

```
    2 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 50/80/110 ms

[R1]ping ipv6 -c 2 2001:FACE::3
  PING 2001:FACE::3 : 56   data bytes, press CTRL_C to break
    Reply from 2001:FACE::3
    bytes=56 Sequence=1 hop limit=64   time = 100 ms
    Reply from 2001:FACE::3
    bytes=56 Sequence=2 hop limit=64   time = 60 ms

  --- 2001:FACE::3 ping statistics ---
    2 packet(s) transmitted
    2 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 60/80/100 ms

[R1]
```

3.2 Configuring OSPFv3

Configure OSPFv3 on R1, R2, and R3. Start the OSPFv3 process on R1, R2, and R3, and specify the router IDs of R1, R2, and R3. Enable the OSPFv3 process on interfaces and specify the area to which the interfaces belong.

```
[R1]ospfv3 1
[R1-ospfv3-1]router-id 1.1.1.1
[R1-ospfv3-1]quit
[R1]
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ospfv3 1 area 0
[R1-GigabitEthernet0/0/0]quit
[R1]
[R1]interface loopback 0
[R1-LoopBack0]ospfv3 1 area 0
[R1-LoopBack0]quit
[R1]

[R2]ospfv3 1
[R2-ospfv3-1]router-id 2.2.2.2
[R2-ospfv3-1]quit
[R2]
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]ospfv3 1 area 0
[R2-GigabitEthernet0/0/0]quit
[R2]
[R2]interface loopback 0
[R2-LoopBack0]ospfv3 1 area 0
[R2-LoopBack0]quit
[R2]

[R3]ospfv3 1
```

```
[R3-ospfv3-1]router-id 3.3.3.3
[R3-ospfv3-1]quit
[R3]
[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]ospfv3 1 area 0
[R3-GigabitEthernet0/0/0]quit
[R3]
[R3]interface loopback 0
[R3-LoopBack0]ospfv3 1 area 0
[R3-LoopBack0]quit
[R3]
```

3.3 Checking OSPFv3 Neighbor Relationships

Check OSPFv3 neighbor relationships on R1, R2, and R3.

```
[R1]display ospfv3 peer
OSPFv3 Process (1)
OSPFv3 Area (0.0.0.0)
Neighbor ID     Pri   State          Dead Time Interface        Instance ID
2.2.2.2           1   Full/Backup    00:00:32  GE0/0/0                    0
3.3.3.3           1   Full/DR        00:00:32  GE0/0/0                    0

[R2]display ospfv3 peer
OSPFv3 Process (1)
OSPFv3 Area (0.0.0.0)
Neighbor ID     Pri   State          Dead Time Interface        Instance ID
1.1.1.1           1   Full/DROther   00:00:37  GE0/0/0                    0
3.3.3.3           1   Full/DR        00:00:39  GE0/0/0                    0

[R3]display ospfv3 peer
OSPFv3 Process (1)
OSPFv3 Area (0.0.0.0)
Neighbor ID     Pri   State          Dead Time Interface        Instance ID
1.1.1.1           1   Full/DROther   00:00:33  GE0/0/0                    0
2.2.2.2           1   Full/Backup    00:00:34  GE0/0/0                    0
```

The command output shows that the neighbor relationships are in the **Full** state and 3.3.3.3 is the DR.

3.4 Checking the OSPFv3 Routing Tables

```
<R1>display ospfv3 routing

Codes : E2 - Type 2 External, E1 - Type 1 External, IA - Inter-Area,
        N - NSSA, U - Uninstalled

OSPFv3 Process (1)
     Destination                                              Metric
       Next-hop
     2001:1::A/128                                                 0
```

```
          directly connected, LoopBack0
        2001:2::B/128                                                   1
          via FE80::2, GigabitEthernet0/0/0
        2001:3::C/128                                                   1
          via FE80::3, GigabitEthernet0/0/0
        2001:FACE::/64                                                  1
          directly connected, GigabitEthernet0/0/0
<R1>

<R2>display ospfv3 routing

Codes : E2 - Type 2 External, E1 - Type 1 External, IA - Inter-Area,
        N - NSSA, U - Uninstalled

OSPFv3 Process (1)
        Destination                                                 Metric
          Next-hop
        2001:1::A/128                                                   1
          via FE80::1, GigabitEthernet0/0/0
        2001:2::B/128                                                   0
          directly connected, LoopBack0
        2001:3::C/128                                                   1
          via FE80::3, GigabitEthernet0/0/0
        2001:FACE::/64                                                  1
          directly connected, GigabitEthernet0/0/0
<R2>

<R3>display ospfv3 routing

Codes : E2 - Type 2 External, E1 - Type 1 External, IA - Inter-Area,
        N - NSSA, U - Uninstalled

OSPFv3 Process (1)
        Destination                                                 Metric
          Next-hop
        2001:1::A/128                                                   1
          via FE80::1, GigabitEthernet0/0/0
        2001:2::B/128                                                   1
          via FE80::2, GigabitEthernet0/0/0
        2001:3::C/128                                                   0
          directly connected, LoopBack0
        2001:FACE::/64                                                  1
          directly connected, GigabitEthernet0/0/0
<R3>
```

3.5 Testing Network Connectivity

On R1, ping the IPv6 addresses of loopback0 interfaces of R2 and R3.

```
<R1>ping ipv6 -c 2 2001:2::B
  PING 2001:2::B : 56    data bytes, press CTRL_C to break
```

```
    Reply from 2001:2::B
    bytes=56 Sequence=1 hop limit=64   time = 50 ms
    Reply from 2001:2::B
    bytes=56 Sequence=2 hop limit=64   time = 50 ms

  --- 2001:2::B ping statistics ---
    2 packet(s) transmitted
    2 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 50/50/50 ms

<R1>ping ipv6 -c 2 2001:3::C
  PING 2001:3::C : 56   data bytes, press CTRL_C to break
    Reply from 2001:3::C
    bytes=56 Sequence=1 hop limit=64   time = 50 ms
    Reply from 2001:3::C
    bytes=56 Sequence=2 hop limit=64   time = 60 ms

  --- 2001:3::C ping statistics ---
    2 packet(s) transmitted
    2 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 50/55/60 ms

<R1>
```

4 Verification

Verification has been performed during the lab.

5 Quiz

1. (Multiple-answer question) Which of the following types of LSAs belong to OSPFv3? (　ABD　)
   A. Router-LSA
   B. Network-LSA
   C. Network-Summary-LSA
   D. Link-LSA
2. By default, the OSPFv3 network type of an Ethernet interface is broadcast, and the OSPFv3 network type of a serial interface (when PPP or HDLC is used for encapsulation) is P2P.
3. What is the command for checking OSPFv3 neighbor relationships?